DATABASE DESIGN

Core Database Concepts

A complete chapter review with practical examples.

Tables & keys, NULL, normalization
Integrity, metadata, physical access

Practical Web Database Design

Chapter 2 - Core Database Concepts

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02

What Is a Data Model?

GENERAL FRAMEWORK

Rules for structuring and manipulating data.

Example: the Relational Model.

SPECIFIC DESIGN

Structures for one particular system.

Example: a library database schema.

Framework = building regulations. Specific design = plans for one building.

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Three Aspects of a Data Model

STRUCTURE

How data is represented.

Tables, rows, and columns.

INTEGRITY

Rules for valid and consistent data.

Keys and permitted values.

OPERATIONS

Ways to retrieve and change data.

Read, insert, update, delete.

This chapter covers structure and integrity. SQL operations follow in Chapter 3.

04 Physical Data Independence

APPLICATION

Queries use tables and columns.

→

DBMS

Maps logical structures to storage.

→

STORAGE

Files, pages, and record locations.

```
SELECT Name FROM EMPLOYEE;
```

Physical storage may change while the logical interface remains stable.

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The Relational Model

ORIGIN

Edgar F. Codd introduced the model in 1970.

Foundations: set theory and predicate logic.

PRACTICAL ROLE

A logical framework for organizing related facts.

Examples: Oracle, DB2, SQL Server, MySQL, PostgreSQL.

RDBMS means Relational Database Management System.

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Tables, Rows & Columns

EmployeeNo	Name	Position	Salary
1	Anna Horton	Programmer	22000
2	Jane Smith	Manager	35000

EMPLOYEE: a table of employee facts

One row = one employee. One column = one property.

A relationship links facts, such as a sale and its customer.

Relation = table. Tuple = row. Attribute = column.

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Rules of Relational Tables

VALUES & IDENTITY

Named columns have permitted values.

Each cell holds one value in the chapter's 1NF treatment.

Relations contain no identical rows.

ORDER & REPRESENTATION

Whole rows and columns can change position without changing facts.

SQL result order requires ORDER BY.

Logical tables need not map to single files.

A spreadsheet-like appearance does not make a relation a spreadsheet.

08 Common Data Types

Category	Examples	Typical content
Numeric	INTEGER, DECIMAL	Counts and prices
Text	CHAR, VARCHAR	Names and addresses
Temporal	DATE, TIME, INTERVAL	Dates, times, durations
Boolean	TRUE / FALSE	Active status
Large objects	BLOB / CLOB	Binary files / large text

Data types define representation and supported operations. Product names vary.

09 Domains: Which Values Are Allowed?

Column	Type	Domain
Age	Integer	0 to 120
Price	Decimal	Zero or greater
Status	Text	Pending / Approved / Rejected

```
Age INTEGER CHECK (Age BETWEEN 0 AND 120)
```

Add NOT NULL separately if the value is mandatory.

A type describes the kind of value. A domain restricts the permitted values.

Missing Information: NULL

UNKNOWN / NOT SUPPLIED

The customer did not provide an email address.

A value may exist, but we do not know it.

INAPPLICABLE

An employee does not receive commission.

A commission rate does not apply.

NULL marks missing information. It does not identify the reason for its absence.

11 NULL in Comparisons & Expressions

Expression	Result
NULL + 2000	NULL
NULL > 2000	UNKNOWN
NULL = NULL	UNKNOWN

THREE-VALUED LOGIC

TRUE
FALSE
UNKNOWN

```
WHERE CreditLimit IS NULL
WHERE CreditLimit IS NOT NULL
```

WHERE retains TRUE rows. FALSE and UNKNOWN do not pass the filter.

NULL & Aggregate Functions

Inputs: 100, 200, NULL

Expression	Result
SUM(Amount)	300
AVG(Amount)	150
COUNT(Amount)	2
COUNT(*)	3

NO KNOWN INPUTS

All NULL, or no input rows:

SUM = NULL
AVG = NULL
COUNT(Amount) = 0

AVG uses the known inputs. COUNT(*) includes rows with NULL values.

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NULL, Defaults & Required Values

NULL & NOT NULL

Allow NULL when absence is meaningful.

Use NOT NULL when the value is mandatory.

A tool may display NULL as a blank.

DEFAULT VALUES

Supply a meaningful predefined value when an insert omits a column.

An explicit NULL normally remains NULL.

Unknown price does not mean zero.

A default is appropriate only when its business meaning is valid.

Primary Keys: Row Identity

ProductNo (PK)	Description	Price
P1	Keyboard	25.00
P2	Mouse	15.00

GOOD IDENTIFIER

ProductNo is a reliable unique code.

WEAK IDENTIFIER

Description can repeat or change.

A primary key is unique, contains no NULL components, and should remain stable.

Composite, Candidate & Surrogate Keys

OrderNo	ProductNo	Qty
100	P1	2
100	P2	5
101	P1	3

Composite key: (OrderNo, ProductNo)

KEY CHOICES

Candidate: minimal unique identifier.

Primary: selected candidate.
Alternate: another candidate.

Surrogate: artificial identifier.

Primary key describes a role. Surrogate key describes an artificial origin.

Foreign Keys: Related Tables

CustomerNo (PK)	Name
C1	Amelia
C2	Richard



SaleNo (PK)	CustomerNo (FK)
100	C1
101	C1

CUSTOMER: parent

SALE: child

ProductNo in SALE can also reference PRODUCT.
A table can have several foreign keys.

Child tables hold foreign keys that reference eligible unique keys in parent tables.

Foreign-Key Rules & JOINs

CustomerNo in SALE	Outcome
C1 (exists)	Valid; may repeat
C9 (absent)	Reject
NULL	Valid if permitted

PARENT ROWS

A customer may exist without sales.

Matching child rows are not required.

```
SELECT s.SaleNo, c.Name
FROM SALE s JOIN CUSTOMER c
  ON s.CustomerNo = c.CustomerNo;
```

Foreign keys enforce references. JOINs combine related information for queries.

Redundancy Creates Anomalies

SaleNo	Customer	Address
100	C1 - Amelia	Cairo
101	C1 - Amelia	Cairo

UPDATE

Change every address copy.

DELETE

Lose customer details with the last sale.

INSERT

Cannot naturally add a customer before a sale.

Repeated customer details create update, delete, and insert problems.

Normalization & Dependencies

```
ProductNo  -> Description, Price
CustomerNo -> Name, Address
SaleNo     -> SaleDate, CustomerNo, ProductNo, Qty
```

PRODUCT

Product facts

CUSTOMER

Customer facts

SALE

Sale facts + references

Normalization separates facts according to what determines them.

First Normal Form: Single Values

BEFORE: repeating values in a cell

OrderNo	Products
100	P1, P2

AFTER: one product per row

OrderNo	ProductNo
100	P1
100	P2

Each cell contains one value.
Use (OrderNo, ProductNo) as the key when each product occurs once per order.

1NF removes lists and repeating groups. Redundancy can still remain.

21 Second Normal Form: The Whole Key

BEFORE: key = (OrderNo, ProductNo)

OrderNo	ProductNo	Description	Qty
100	P1	Keyboard	2
101	P1	Keyboard	3

ProductNo determines Description.

AFTER: separate product facts

ProductNo (PK)	Description
P1	Keyboard

OrderNo	ProductNo	Qty
100	P1	2
101	P1	3

2NF requires 1NF and no partial candidate-key dependency of non-key attributes.

Third Normal Form: Transitive Dependency

```
ProductNo -> SupplierCode -> SupplierName, SupplierAddress
```

ProductNo	SupplierCode	SupplierName	Address
P1	S1	Tech Supply	Cairo
P2	S1	Tech Supply	Cairo

AFTER: supplier details have one home

PRODUCT(ProductNo, Description, Price, SupplierCode)
SUPPLIER(SupplierCode, SupplierName, SupplierAddress)

Keep supplier facts in SUPPLIER and retain SupplierCode as a product reference.

Fourth Normal Form: Independent Facts

Independent: 2 addresses x 2 phones = 4 combinations

CustomerNo	Address	Phone
C1	Cairo	111
C1	Cairo	222
C1	Giza	111
C1	Giza	222

AFTER: TWO TABLES

CUSTOMER_ADDRESS
(CustomerNo, Address)

CUSTOMER_PHONE
(CustomerNo, Phone)

Two rows in each table.

4NF addresses independent multivalued facts, even when every cell has one value.

Higher Forms & Denormalization

HIGHER NORMAL FORMS

5NF addresses join dependencies.

3NF or BCNF is a common practical target.

Use higher forms when dependencies require them.

DENORMALIZATION

Deliberately duplicate selected facts for a particular read workload.

Operational systems often favor normalization.

Warehouses often favor analytical reads.

More joins do not automatically mean slower queries. Measure the workload.

Three Categories of Data Integrity

ENTITY

Primary keys are unique.

No primary-key component is NULL.

REFERENTIAL

References match existing parent keys.

NULL is possible when permitted.

DOMAIN

Values obey the allowed domain.

Types, ranges, and permitted categories.

Integrity keeps stored information valid and internally consistent.

Business Rules: Valid vs Correct

BUSINESS RULE

CreditLimit must be between
0 and 5000.

FACTUAL CHECK

Actual agreed limit: 1000.
Entered limit: 4000.

Valid range, wrong fact.

Browser: immediate feedback.
Server / database: enforce suitable rules beyond the browser.
Legacy data: identify violations and plan how to handle them.

Constraints validate rules. Verification checks whether the facts are actually correct.

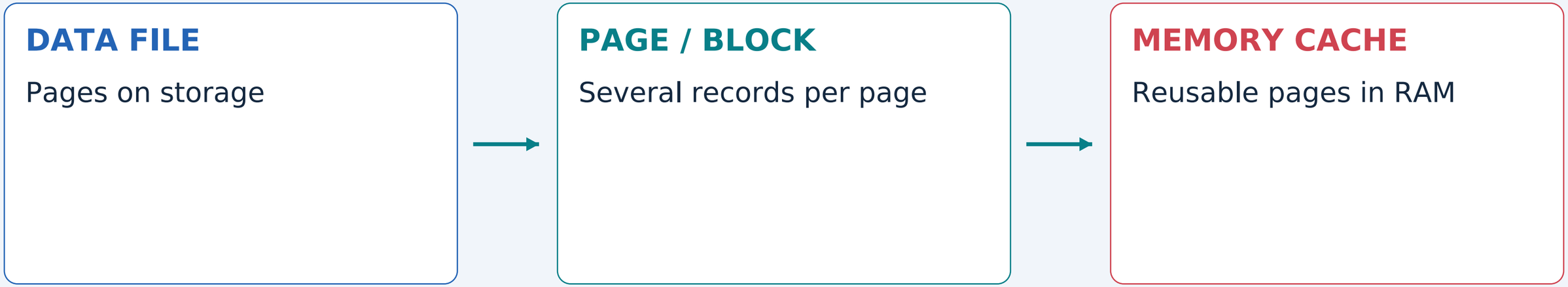
Metadata & the Data Dictionary

Business data	Metadata
Anna earns 22000	Salary is a decimal column
Customer C1 lives in Cairo	Address is a text column
Employee number is 15	EmployeeNo is the primary key

Catalog: definitions, constraints, privileges, indexes, statistics.
The DBMS uses these details to validate operations and plan access.

The catalog stores definitions the DBMS uses to understand and manage the database.

Pages, Blocks & Caching



Not cached: read the page into RAM.
Already cached: reuse it without another storage read.
Then locate the required record inside the page.

A page access causes a storage read only when the required page is not already cached.

Sequential Access: Scanning Records



Best case	Average case	Worst case
1 record	About n / 2 records	n records

Average assumes one existing match, equally likely at any position.

Records examined and pages read are different measures.

Indexed Access: A Separate Directory

Index key	Data location
Chen	Page 2
Smith	Page 8
Tudor	Page 4



DATA PAGE 4

Record for Tudor

Unique index: restrict duplicate keys.
Non-unique index: permit duplicate keys.

An index narrows the search before the DBMS accesses the associated data.

B-tree Search: Following a Path

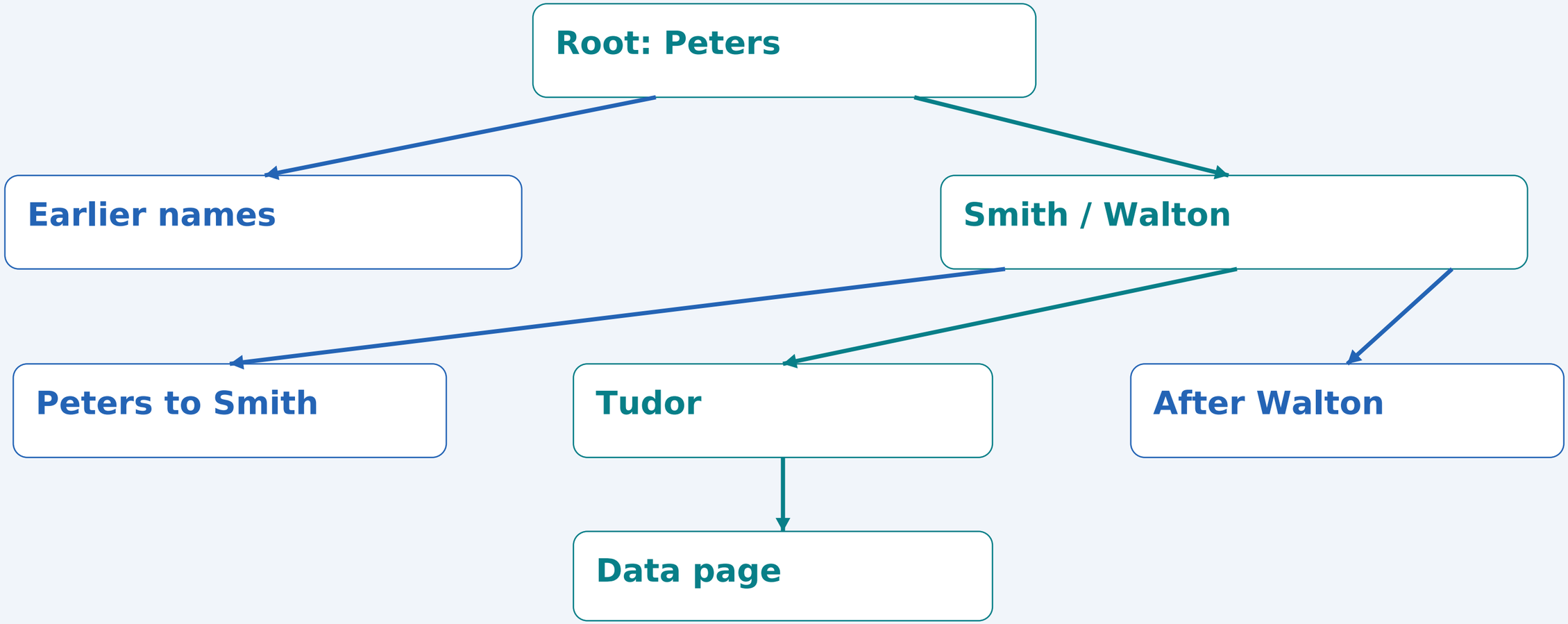


Illustration: three index pages plus one data page, if all four are uncached.

Range Retrieval & Index Costs

ORDERED SEARCHES

Find the first key in a range, then continue through matching entries.

Example: surnames starting with C.

Table pages need not share index order.

MAINTENANCE

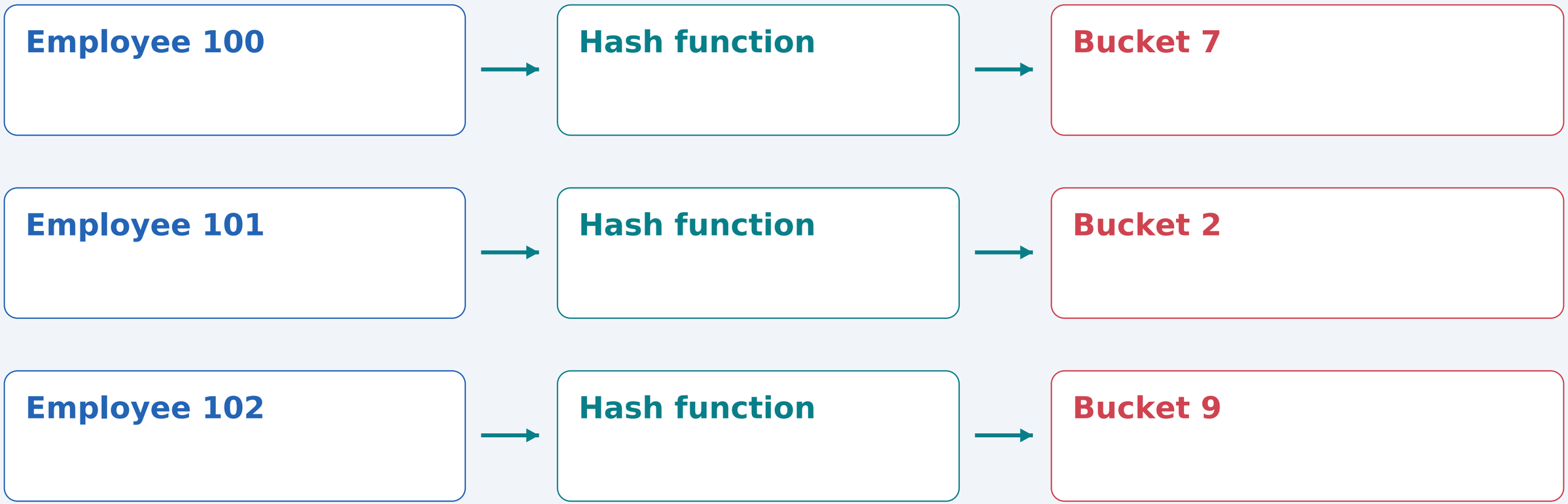
Indexes consume space.

Inserts, deletes, and indexed changes require maintenance.

Full pages can split.

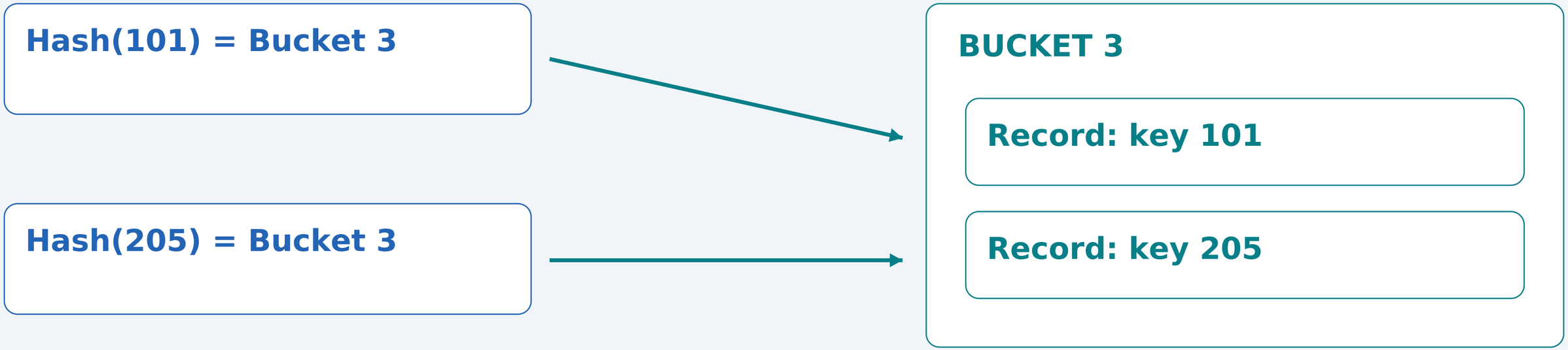
Indexes help suitable reads, but add write work. A small-table scan can be cheaper.

Direct Access Through Hashing



Hashing selects a bucket. The DBMS then locates the matching entry inside it.

Hash Buckets, Collisions & Overflow



Separate record slots in a shared bucket.
A bucket can span one or more pages.

A collision shares a destination. Overflow means the bucket needs more capacity.

Choosing an Access Method

Method	Useful workload	Trade-off
Sequential	Small table / many rows	May scan irrelevant data
Tree index	Exact matches + ranges	Space + write maintenance
Hashing	Exact-match lookups	No ordered range traversal

No universal winner.
Consider query selectivity, caching, updates, and available structures.

One query can use different access methods for different tables or plan steps.

Putting It All Together

CUSTOMER

CustomerNo (PK)
Name, Address

SALE

SaleNo (PK)
CustomerNo (FK)
ProductNo (FK)
Qty, Amount

PRODUCT

ProductNo (PK)
Description, Price

Keys connect facts. Normalization limits redundancy.
Integrity protects rules. Metadata describes structures.
Pages, caching, and access methods support retrieval.

KEY TAKEAWAY: Organize facts clearly, enforce their rules, and match access to the workload.

Created with AI assistance. Reviewed and presented by Abdalla Fadl.